Chi-Square Test for Independence

Statistics Worksheet · Grade 11–College

Name: ________________________________ Date: __________________

Learning Objectives

- Identify when a chi-square test for independence is appropriate and state correct null and alternative hypotheses
- Compute expected counts, the chi-square test statistic, and degrees of freedom from a two-way contingency table
- Verify conditions for the chi-square test and draw conclusions based on p-value comparisons

Problems

1. A researcher surveys a random sample of adults and records both their political party (Democrat, Republican, Independent) and their opinion on a new tax policy (Favor, Oppose). Which type of chi-square test should be used, and why?

2. A survey on handgun ban opinion cross-classified 1,201 adults by education level and response (Yes or No). State the null and alternative hypotheses for the chi-square test for independence.

3. A two-way table has 3 rows and 4 columns. Calculate the degrees of freedom for a chi-square test of independence.

$$df = (r - 1)(c - 1)$$

4. The observed two-way table below shows survey counts for opinion on a handgun ban by education level. Complete the row totals, column totals, and grand total.

Education Level	Yes	No	Row Total

5. Using the completed table from the previous problem, verify the three conditions required before performing a chi-square test for independence: randomness, the 10% condition, and large counts (expected counts ≥ 5).

6. Calculate the expected count for the cell corresponding to 'Less than HS' education and 'Yes' response, using the totals: row total = 95, column total = 446, and grand total = 1201.

$$E = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Grand Total}}$$

7. The observed count for a cell is 42 and its expected count is 35.28. Calculate the chi-square contribution for this single cell.

$$\chi^2_{\text{cell}} = \frac{(O - E)^2}{E}$$

8. Using the observed and expected counts table below, calculate the total chi-square test statistic by summing all cell contributions.

Cell	O	E	$(O-E)^2/E$

9. The chi-square test statistic for the handgun ban survey is approximately 26.78 with 4 degrees of freedom. The p-value is less than 0.0001. Using a significance level of 5%, state the conclusion of the hypothesis test in context.

$$\alpha = 0.05, \quad \chi^2 \approx 26.78, \quad df = 4, \quad p < 0.0001$$

10. A different study surveys two separate groups — one group of gun owners and one group of non-gun-owners — and records whether each person supports or opposes stricter background checks. A student claims this should also use a chi-square test for independence. Is the student correct? Identify the correct test and explain the key distinction.

Chi-Square Test for Independence — Answer Key

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Answer Key

1. Answer: Chi-square test for independence; one random sample is classified according to two categorical variables.

- Identify the sampling design: one single random sample of adults.
- Each individual is classified by two variables: political party and tax opinion.
- When a single sample is cross-classified by two categorical variables, the appropriate test is the chi-square test for independence.

2. Answer: H0: There is no association between education level and opinion about a handgun ban in the adult population. Ha: There is an association between education level and opinion about a handgun ban in the adult population.

- The null hypothesis always states independence (no association) between the two categorical variables.
- The alternative hypothesis states that an association exists.
- H0: Education level and handgun ban opinion are independent in the adult population.
- Ha: Education level and handgun ban opinion are associated in the adult population.

3. Answer: df = 6

- The formula for degrees of freedom is df = (number of rows – 1)(number of columns – 1).
- df = (3 – 1)(4 – 1) = (2)(3) = 6.

4. Answer: See completed table

Education Level	Yes	No	Row Total
Less than HS	42	53	95
HS Graduate	188	230	418
Some College	90	159	249
College Grad	70	181	251

Education Level	Yes	No	Row Total
Postgraduate	56	132	188
Column Total	446	755	1201

5. Answer: All three conditions are satisfied: random sample stated, n = 1201 is less than 10% of all adults, and all expected counts exceed 5.

- Randomness: The problem states a random sample of adults was selected — satisfied.
- 10% condition: The total population of adults is far greater than 10 × 1201 = 12,010 — satisfied.
- Large counts: Each expected count = (row total × column total) / grand total must be ≥ 5. The smallest row total is 95; its smallest expected count = (95 × 446) / 1201 ≈ 35.3 ≥ 5 — satisfied.

6. Answer: E ≈ 35.30

- Apply the expected count formula: E = (row total × column total) / grand total.
- E = (95 × 446) / 1201 = 42,370 / 1201 ≈ 35.28.
- Round to two decimal places: E ≈ 35.28 (approximately 35.30).

7. Answer: ≈ 1.28

- O = 42, E = 35.28.
- $(O – E)^2 = (42 – 35.28)^2 = (6.72)^2 = 45.1584$.
- Divide by E: 45.1584 / 35.28 ≈ 1.28.

8. Answer: $\chi^2 \approx 26.78$

Cell	O	E	$(O-E)^2/E$
Less than HS / Yes	42	35.28	1.28
Less than HS / No	53	59.72	0.76
HS Grad / Yes	188	155.16	6.95

Cell	O	E	(O-E)²/E
HS Grad / No	230	262.84	4.10
Some College / Yes	90	92.45	0.06
Some College / No	159	156.55	0.04
College Grad / Yes	70	93.21	5.79
College Grad / No	181	157.79	3.42
Postgrad / Yes	56	69.87	2.75
Postgrad / No	132	118.13	1.63
Total			26.78

9. Answer: Reject H0. There is convincing evidence of an association between education level and opinion about a handgun ban in the adult population.

- Compare p-value to significance level: $p < 0.0001 < 0.05 = \alpha$.
- Since $p < \alpha$, reject the null hypothesis.
- Conclusion: The data provide convincing evidence that education level and opinion about a handgun ban are associated in the adult population.

10. Answer: No. This is a chi-square test for homogeneity because two separate samples (gun owners and non-gun-owners) were drawn independently, not one single sample classified by two variables.

- Test for independence: one random sample is collected, and each individual is classified according to two categorical variables.
- Test for homogeneity: two or more separate random samples are drawn from distinct populations and compared on one categorical variable.

• Here, two distinct groups (gun owners and non-gun-owners) were sampled separately — this matches the test for homogeneity.
• The student is incorrect; the appropriate test is the chi-square test for homogeneity.
• Note: the calculation procedure is identical for both tests — only the study design and hypotheses differ.